

Hassan Khazal

Gameplay Programmer and Technical Designer

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UK hybrid and on-site roles | Remote opportunities worldwide

Profile

Technical Designer and Gameplay Programmer with over 10 years of professional programming experience, including 6+ years in game development. I bridge creative design and engineering to build engaging gameplay across Unreal Engine and Unity, from AI, combat and physics to VR and multiplayer. My experience includes leading multiplayer implementation, integrating online services, and developing reusable gameplay features and editor tools. I enjoy collaborating with designers, solving gameplay challenges and creating experiences that feel rewarding to play.

Technical skills

C++, C#, Blueprints | Unreal Engine 4/5, Unity | Behaviour Trees, EQS | Physics and VR interaction | Replication, Epic Online Services | UMG, Slate, editor tools | Git, Perforce, SVN | Jira, Confluence, Notion

Professional experience

Multiplayer, Gameplay & Systems Programmer - Freelance

Simulated Minds | Sep 2024 - Present

- Led Plastic Battlegrounds VR's multiplayer conversion, covering replicated gameplay, VR interaction, EOS lobbies and sessions, seamless travel, Player Data Storage, and proximity/team voice.
- Built developer-configured spawning menus with entity settings and customisations, plus persistent unlocks that save and restore the player's available spawnables.
- Reworked teams around reusable components, data tables and interfaces; developed replicated cosmetics and store integration.
- Created custom wheel physics and tank steering, with slope detection and optional gravity compensation for steep inclines. Developed a physical catapult for Cry of Athena using physics constraints.
- Directed contractors and established coding guidance and documentation; built editor tools to support designer workflows.

Game Programmer

Ringzero Game Studio | May 2023 - May 2024

- Led story-mode implementation for a UE5 hospital/surgery simulation; developed interactive workstations, room upgrades and designer-editable comic-style cutscenes using UMG and Slate.
- Used Unreal Insights, World Partition and Data Layers for profiling and optimisation. Coordinated implementation with designers and programmers, and supported junior onboarding.

Unreal Engine Consultant

Simulated Minds | Cry of Athena | Apr 2022 - Jan 2024

- Implemented AI hiding, targeting, pursuit and vehicle movement using Behaviour Trees and EQS; developed two-, four- and six-plus-wheeled vehicles.
- Built C++ save/load for player-created buildings, bullet penetration and deflection, and editor tools for sandbox content creation.

Gameplay Programmer

RobocomVR | Pixoul Planet | Aug 2020 - Jan 2023

- Contributed to ten shipped Unity location-based VR games, creating humanoid enemy AI, two-axis turrets and spiders. Proposed and implemented a spider attack that jumps onto the player's face.
- Built varied enemy cover interactions, including ducking, vaulting and sliding. Created Battle Rush's sequential train boss encounter, AI pursuit and vehicle-bashing mechanics.
- Led development of four-player Pixoul Recon. Proposed room-based combat linked by platform travel to address poor visibility at long range while retaining the environment's scenery.
- Integrated ProTube VR guns and simulator machines through APIs and network connections. Implemented VR player IK, scale calibration, weapon recoil and attachments.

Web Developer

LCIS | Feb 2015 - Nov 2020

- Developed and maintained four news and entertainment websites, covering deployment, security updates and collaboration with designers and backend developers.

Independent projects

- HK Diegetic Dynamic Map: C++ Unreal plugin with scene-capture maps, configurable points of interest, spline routes and optional occlusion. Fab release planned.
- C++ Actor Pooling Plugin: work in progress with configurable actor pre-spawning. Replication and a Fab release are planned.

Education and professional development

Unreal Engine 5 Technical Art Skills Bootcamp | INFINITY27 | Jul 2026

Completed with a pass.

Sumo Group Evolve | Jun 2024

Completed the emerging leaders programme for the video games industry.

Management Informatics Vocational Degree | Lebanese International Institute

Oct 2013 - Jun 2014 | Programming and software design

Languages: English and Arabic